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Multiplication polynomials for elliptic curves over finite local rings

Polynomial Solving Reading Seminar

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0 Outline

1. Notation and Standard Results
2. Addition Law
3. Points at infinity
4. Multiplication Polynomials
5. Consequences

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1 Outline

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$$y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3,$$

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  for $a_i \in \mathcal{R}$ and $\Delta_E \in \mathcal{R}^*$;
- $\mathcal{O} = (0 : 1 : 0)$;
- $-(X : Y : Z) = (X : -Y - a_1X - a_3Z : Z)$;

1 Notation and Standard Results

- The *standard form* for $P \in E$ is

$$\begin{cases} (X \cdot Z^{-1} : Y \cdot Z^{-1} : 1), & \text{if } Z \in \mathcal{R}^*, \\ (X \cdot Y^{-1} : 1 : Z \cdot Y^{-1}), & \text{otherwise.} \end{cases}$$

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- $E^\infty = \pi^{-1}(\mathcal{O}) < E(\mathcal{R})$ is the *group at infinity*.

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2 Addition Law

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- every addition law works in $\mathcal{P}^2$ except from a line;
- on local ring one addition law always works;
- $+_{(0:1:0)}$ and $+_{(0:0:1)}$ form a *complete set of addition laws*.

2 Addition Law

Proposition

For $i \in \{1, 2\}$, let $P_i = (X_i : Y_i : Z_i) \in \mathcal{P}^2(\mathcal{R})$ be two projective points. Let also $(X_3, Y_3, Z_3) = P_1 +_{(0:1:0)} P_2$, and

$$g1 = X_2(a_1X_1 + a_3Z_1 + Y_1) + X_1Y_2,$$
$$g2 = Z_2(a_1X_1 + a_3Z_1 + Y_1) + Z_1Y_2.$$

Then

$$X_3 = g_1H_1 + g_2H_2,$$
$$Z_3 = g_1H_3 + g_2H_4,$$
$$Y_3 = H_1H_4 - H_2H_3.$$

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- notation: $P = (P_x : 1 : P_z)$.

Remark

In general $P_1 + P_2$ *is not* in standard form.

3 Points at infinity

Proposition

There is a polynomial $\mathtt{f} \in \mathcal{R}[x]$ of degree strictly lower than the nilpotence degree of $\mathcal{R}$, such that for every $P \in E^\infty$ we have $P = (P_x : 1 : \mathtt{f}(P_x))$. Moreover, $x^3 | \mathtt{f}(x)$.

Proof

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Proof

In E^∞ we have $y = 1$. From Weierstrass equation we get

$$z = f(x, z) = x^3 - a_1 xz + a_2 x^2 z - a_3 z^2 + a_4 x z^2 + a_6 z^3$$

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3 Points at infinity

Proposition (semi-linearity of the sum)

Let $P_1 = (X_1 : 1 : Z_1), P_2 = (X_2 : 1 : Z_2) \in E^\infty$ be two points and $P_3 = P_1 +_{(0:1:0)} P_2$. Let also $I_P = \langle X_1^2, Z_1 \rangle$. Then

$$\begin{aligned}(P_3)_x \equiv\ & X_1 + X_2 \\ & + (a_1 X_2 - a_2 X_2^2 + 2a_3 Z_2 - 2a_4 X_2 Z_2 - 3a_6 Z_2^2) X_1 \bmod I_P.\end{aligned}$$

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Proof

Direct inspection of addition law.

3 Points at infinity

Corollary

Given any curve E over $\mathcal{R}$ and three points $P, Q, R \in E^\infty$ such that $P + Q = R$, we have

$$R_x \in \langle P_x, Q_x \rangle \subset \mathbb{Z}[a_1, \ldots, a_6][P_x, Q_x].$$

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Remark

For every multiple of $P = (X : 1 : Z)$, we have $(nP)_x \in \langle X \rangle$.

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4 Multiplication Polynomials

Lemma

Given an elliptic curve E over $\mathcal{R}$, for every $n \in \mathbb{N}$ there are uniquely defined coefficients $\psi_1(n), ..., \psi_{k-1}(n) \in \mathcal{R}$ such that for every symbolic $P = (X : 1 : \mathtt{f}(X)) \in E^\infty$ we have

$$(nP)_x = \sum_{i=1}^{k-1} \psi_i(n) X^i.$$

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Definition

For every $1 \leq i \leq k-1$ we define the *i-th multiplication polynomial* ψ_i as the unique function over $\mathbb{N}$ such that $\psi_i(n)$ is the coefficient of X^i in $(nP)_x$.

4 Multiplication Polynomials

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Lemma

It holds

$$\psi_1(n) = n,$$

$$\psi_2(n) = \binom{n}{2} a_1.$$

4 Multiplication Polynomials

Technical Lemma

Let $\mathbb{Z}[a_1, \ldots, a_6][\beta_1, \ldots, \beta_{k-1}]$ be the graded ring of weights $\deg_\beta(\beta_j) = j$. For every $2 \leq i \leq k-1$, there exists

$$g_i \in \langle \beta_1, \ldots, \beta_{i-1} \rangle, \quad \deg_\beta(g_i) = i - 1,$$

such that, if we symbolically compute

$$S = (X : 1 : \mathtt{f}(X)) + \left(\sum_{j=1}^{k} \beta_j X^j : 1 : \mathtt{f} \left(\sum_{j=1}^{k} \beta_j X^j \right) \right), \tag{1}$$

then the coefficient of X^i in S_x is $\beta_i + g_i$.

4 Multiplication Polynomials

Theorem (1)

For every $1 \leq i \leq k-1$, the i-th multiplication polynomial ψ_i is a polynomial in $\mathbb{Q}[a_1, \ldots, a_6][n]$ of degree i in n. Moreover, we have $n | \psi_i(n)$.

4 Multiplication Polynomials

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Theorem (2)

For every $1 \leq i \leq k-1$ we have

$$\Pi(i)\psi_i(n) \in \mathbb{Z}[a_1, ..., a_6][n],$$

where

$$\Pi(i) = \prod_{j=1}^{i} j!.$$

4 Multiplication Polynomials

Corollary

Let p be a prime number. For every exponent $l \geq 1$ and for every $1 \leq i < p$, we have

$$\psi_i(p^l) \equiv 0 \mod p^l.$$

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Proof

By Theorem (2) we have $\Pi(i)\psi_i(p^l) \in \mathbb{Z}[a_1, \ldots, a_6]$. Thanks to Theorem (1) we also have $\Pi(i)\psi_i(p^l) \equiv 0 \mod p^l$. By definition $\Pi(i) \in (\mathbb{Z}/p^l\mathbb{Z})^*$ for every $i < p$, therefore we conclude $\psi_i(p^l) \equiv 0 \mod p^l$.

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- $\pi = (\alpha_x + \beta_x\epsilon : \alpha_y + \beta_y\epsilon : \alpha_z + \beta_z\epsilon) \mapsto (\alpha_x : \alpha_y : \alpha_z)$;
- $0 \to E^\infty(R_k) \overset{i}{\hookrightarrow} E(R_k) \overset{\pi}{\twoheadrightarrow} E(\mathbb{F}_q) \to 0$;
- if $\gcd\left(\#E(\mathbb{F}_q), p\right) = 1$, $E(R_k) \cong E(\mathbb{F}_q) \oplus E^\infty$.

5 Consequences

Corollary

Let $P = (X : 1 : \mathtt{f}(X)) \in E^\infty$ be a point. Then

$$(pP)_x \equiv \psi_p(p) X^p \bmod X^{p+1}.$$

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Proposition

Let E be an elliptic curve over R_k where $k \leq p$. Then we have the group isomorphism

$$E^\infty \cong (\mathbb{F}_p)^{e(k-1)}.$$

5 Consequences

Definition

Let $r \in R_k \backslash \{0\}$. We define its *minimal degree* $\nu(r)$ as the maximal $i \geq 0$ such that $\epsilon^i | r$. We also define $\nu(0) = \infty$. Finally, for every point $P \in E^\infty$, we define $\nu(P) = \nu(P_x)$.

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Proposition

Let $P_1 = (X_1 : 1 : Z_1), P_2 = (X_2 : 1 : Z_2) \in E^\infty$ be two points and $P_3 = P_1 +_{(0:1:0)} P_2$. Let also $m = \min\{\nu(P_1), \nu(P_2)\}$. Then we have

$$(P_3)_x \equiv X_1 + X_2 \bmod \epsilon^{m+1}.$$

5 Consequences

Lemma

Let $P, Q \in E^\infty$ be two points with $\nu(P) \neq \nu(Q)$. Then

$$\nu(P+Q) = \min\{\nu(P), \nu(Q)\}.$$

If instead $\nu(P) = \nu(Q) = m < \infty$, let $c_p, c_q \in \mathbb{F}_q$ be the coefficients such that

$$P_x \equiv c_p \epsilon^m \bmod \epsilon^{m+1} \quad \text{and} \quad Q_x \equiv c_q \epsilon^m \bmod \epsilon^{m+1}.$$

Then we have

$$(P+Q)_x = (c_p + c_q)\epsilon^m \bmod \epsilon^{m+1}.$$

5 Consequences

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Definition

We define the *trajectory* of $P \in E^\infty$ as

$$\mathrm{trj}(P) = \{\nu(nP)\}_{n\in\mathbb{N}} \setminus \{\infty\}.$$

5 Consequences

Proposition

Let

$$l_m = \left\lfloor \log_p \frac{k-1}{m} \right\rfloor + 1.$$

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- $\mathrm{trj}(P) = \{mp^i\}_{i \leq l_m}$;
- $\#\,\mathrm{trj}(P) = l_m$;
- let $\mathcal{A} = \{m : (p, m) = 1 \text{ and } m \leq k-1\}$; then

$$\sum_{m \in \mathcal{A}} l_m = k - 1.$$

5 Consequences

Lemma

Let $P, Q \in E^\infty$ be points with $\mathrm{trj}(P) \cap \mathrm{trj}(Q) = \emptyset$. For every $n, m \in \mathbb{Z}$ we have $nP + mQ = \mathcal{O}$ if and only if $nP = mQ = \mathcal{O}$.

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Proposition

Let $\{\gamma_n\}_{1 \leq n \leq e}$ be an $\mathbb{F}_p$-basis of $\mathbb{F}_q$. For any given $1 \leq m \leq k-1$, the points

$$g_{nm} = \left(\gamma_n \epsilon^m : 1 : \mathtt{f}(\gamma_n \epsilon^m)\right) \in E^\infty$$

are linearly independent. Moreover, the trajectory of every linear combination of the $\{g_{nm}\}_{1 \leq n \leq e}$ lies into $\{mp^j\}_{j < l_m}$.

5 Consequences - Group structure

Theorem

Let E be an elliptic curve over R_k such that $\psi_p(p) \in R_k^*$. Then

$$E^\infty \cong \prod_{\substack{1 \leq m \leq k-1 \\ (m,p)=1}} \left(\mathbb{Z}_{p^l m}\right)^e.$$

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5 Consequences - Group structure

Theorem

Let E be an elliptic curve defined over R_k, such that $\nu(\psi_p(p)) = d > 0$ and

- C1 $\psi_{p^2}(p) \in R_k^*$;
- C2 $\psi_i(p) = 0$ for all $i < p^2$ such that $(i, p) = 1$;
- C3 $\psi_i(p) \in \langle \psi_p(p) \rangle$ for all $i < p^2$ such that $p|i$.

Then there are explicitly computable $\mathcal{A} \subseteq \{1, \ldots, k-1\}$ and $l_m \in \mathbb{N}^*$ for every $m \in \mathcal{A}$, such that we have the group isomorphism

$$E^\infty \cong \prod_{m \in \mathcal{A}} \left(\mathbb{Z}_{p^{l_m}}\right)^e.$$

5 Consequences - ECDLP

From our results, we efficiently recover the discrete logarithm of points in E^∞ over R_k. In fact, we can always write $n = b_0 + b_1 p + \cdots + b_{k-1} p^{k-1}$. Let $m_i = \nu(p^i P)$. Then we have

$$b_i = \left(\left(Q - \sum_{j=1}^{i-1} b_j p^j P \right)_x \bmod \epsilon^{m_i+1} \right) \Big/ \left((p^i P)_x \bmod \epsilon^{m_i+1} \right).$$

The time complexity of this algorithm is logarithmic in the considered parameters. More precisely, it has a time complexity of $\log(p)\log(n)$.

Thanks for the attention.